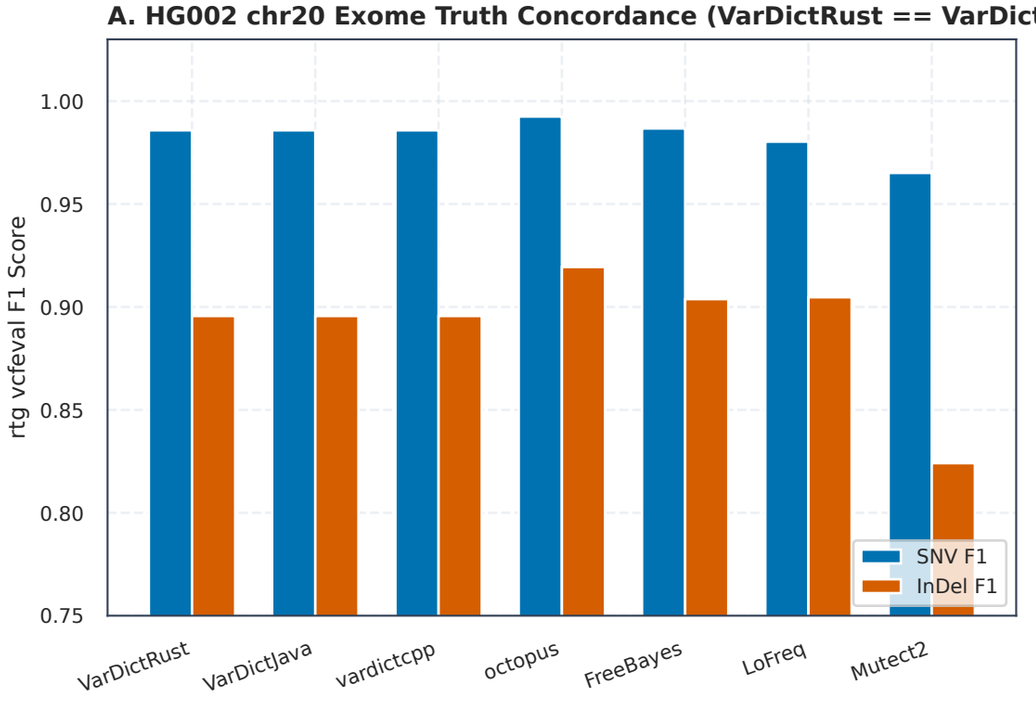
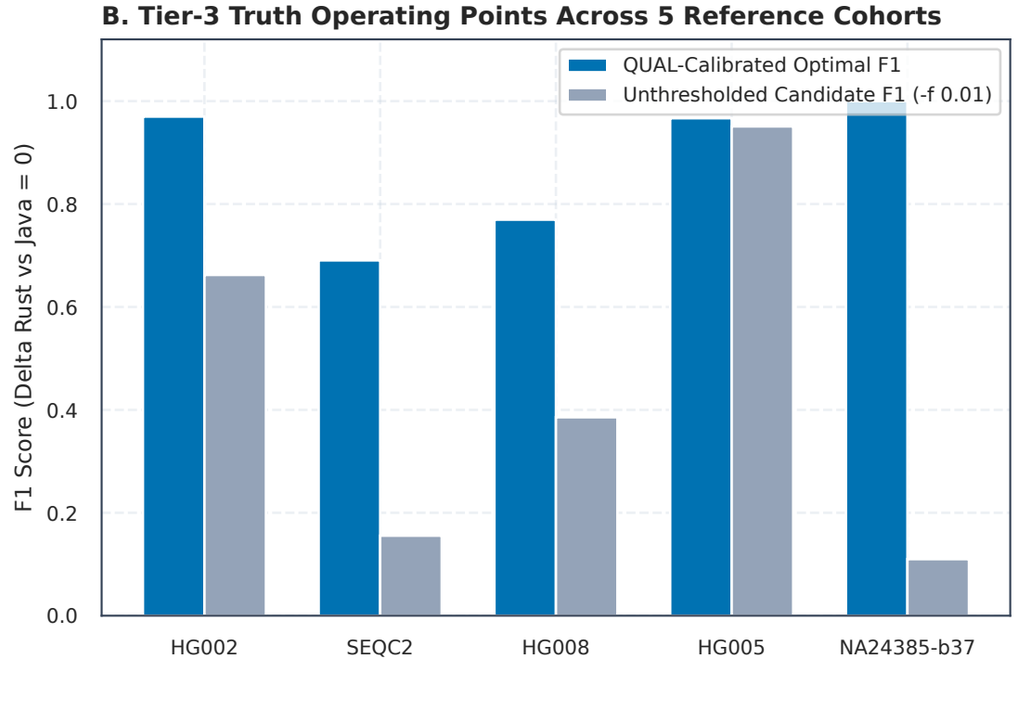
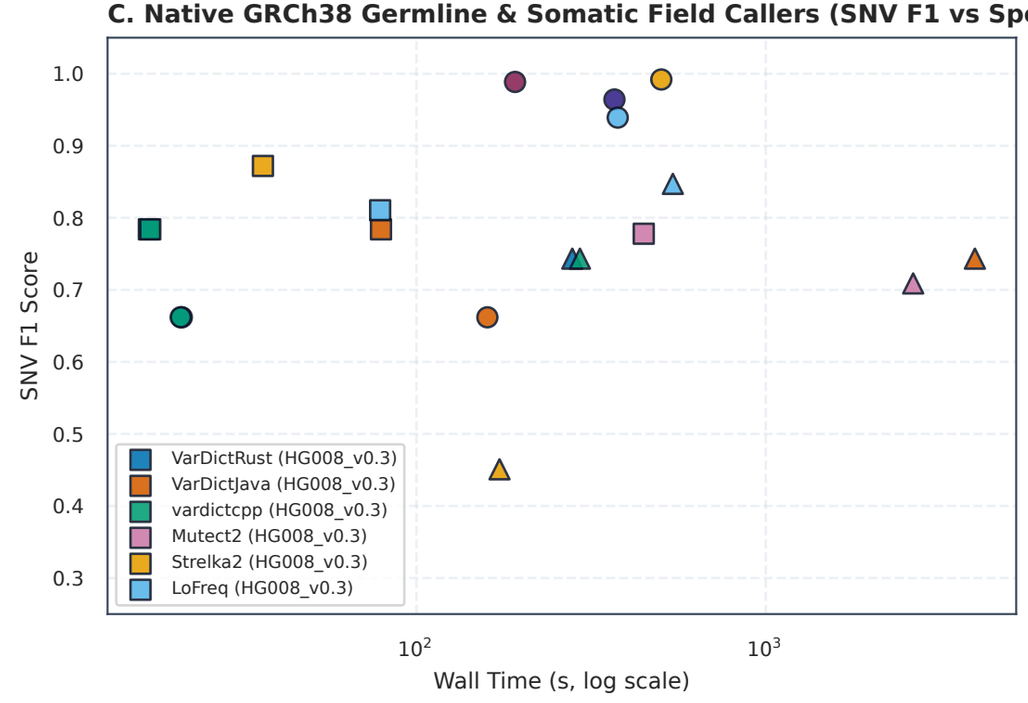
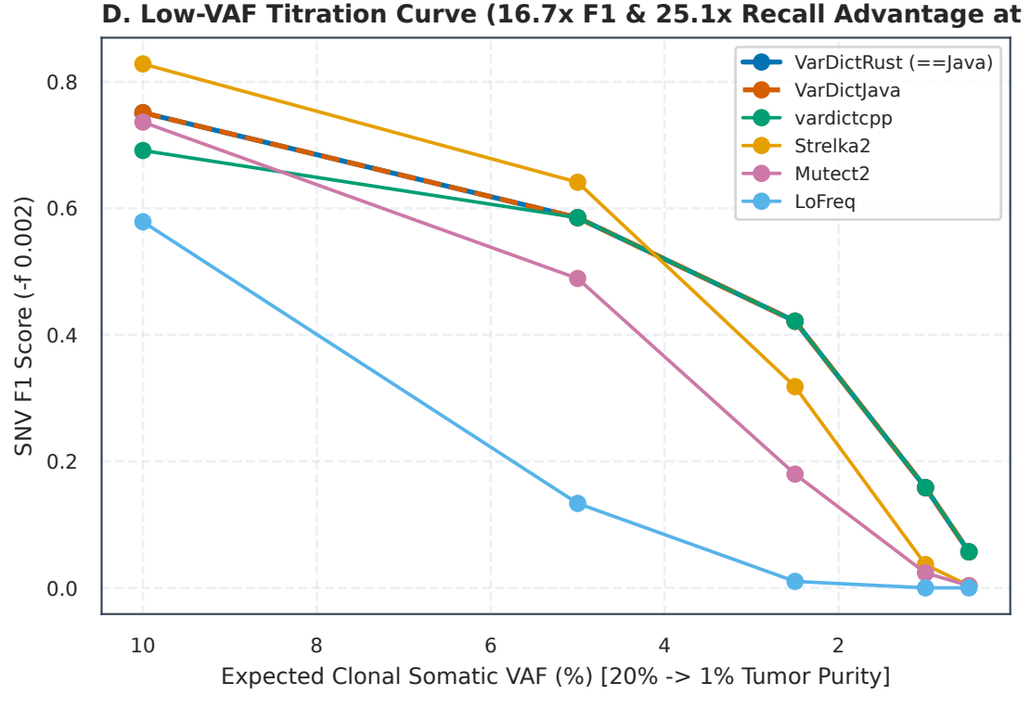

A. HG002 chr20 Exome Truth Concordance (VarDictRust == VarDictJava)
rtg vcfeval F1 Score
1.00
0.95
0.90
0.85
0.80
0.75
SNV F1
InDel F1
VarDictRust
VarDictJava
vardictcpp
octopus
FreeBayes
LoFreq
Mutect2


B. Tier-3 Truth Operating Points Across 5 Reference Cohorts
F1 Score (Delta Rust vs Java = 0)
1.0
0.8
0.6
0.4
0.2
0.0
QUAL-Calibrated Optimal F1
Unthresholded Candidate F1 (-f 0.01)
HG002
SEQC2
HG008
HG005
NA24385-b37


C. Native GRCh38 Germline & Somatic Field Callers (SNV F1 vs Speed)
SNV F1 Score
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
VarDictRust (HG008_v0.3)
VarDictJava (HG008_v0.3)
vardictcpp (HG008_v0.3)
Mutect2 (HG008_v0.3)
Strelka2 (HG008_v0.3)
LoFreq (HG008_v0.3)
10^2
10^3
Wall Time (s, log scale)


D. Low-VAF Titration Curve (16.7x F1 & 25.1x Recall Advantage at 0.5% VAF)
SNV F1 Score (-f 0.002)
0.8
0.6
0.4
0.2
0.0
VarDictRust (==Java)
VarDictJava
vardictcpp
Strelka2
Mutect2
LoFreq
10
8
6
4
2
Expected Clonal Somatic VAF (%) [20% -> 1% Tumor Purity]